
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Linear Algebra Revision 3

Exercise 1. In this exercise, we will prove the following result stated in the class:

Let $\mathcal{S} \subset \mathcal{V} \subseteq \mathbb{R}^n$ be vector subspaces ($\mathcal{S}$ is a subspace of $\mathcal{V}$, which itself is a subspace of $\mathbb{R}^n$). Then, any $\mathbf{v} \in \mathcal{V}$ can be uniquely written as $\mathbf{v} = \mathbf{w} + \mathbf{z}$, where $\mathbf{w} \in \mathcal{S}$ and $\mathbf{z} \perp \mathcal{S}$.

Assume that $\dim(\mathcal{S}) = p$ and $\dim(\mathcal{V}) = p + q$.

- (a) Verify that it is possible to find an orthonormal basis for $\mathcal{S}$, say $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$.
- (b) This basis can be extended into a basis for $\mathcal{V}$, say $\{\mathbf{u}_1, \dots, \mathbf{u}_p, \mathbf{w}_{p+1}, \dots, \mathbf{w}_{p+q}\}$. Apply orthogonalization on $\{\mathbf{u}_1, \dots, \mathbf{u}_p, \mathbf{w}_{p+1}, \dots, \mathbf{w}_{p+q}\}$ to get an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_{p+q}\}$ for $\mathcal{V}$.

Note: Orthogonalization on the vectors $\mathbf{u}_1, \dots, \mathbf{u}_p$ does not change them!

- (c) Verify that the vectors $\mathbf{u}_{p+1}, \dots, \mathbf{u}_{p+q}$ are orthogonal to $\mathcal{S}$.
- (d) Since $\{\mathbf{u}_1, \dots, \mathbf{u}_{p+q}\}$ is a basis for $\mathcal{V}$, we can write any $\mathbf{v} \in \mathcal{V}$ uniquely as $\mathbf{v} = v_1\mathbf{u}_1 + \dots + v_{p+q}\mathbf{u}_{p+q}$. Define $\mathbf{w} = v_1\mathbf{u}_1 + \dots + v_p\mathbf{u}_p$ and $\mathbf{z} = v_{p+1}\mathbf{u}_{p+1} + \dots + v_{p+q}\mathbf{u}_{p+q}$. Check that $\mathbf{w} \in \mathcal{S}$ and $\mathbf{z} \perp \mathcal{S}$.

This proves the result that we wanted to show. Here, we have assumed that $\mathcal{S}$ is a *proper* subspace of $\mathcal{V}$ (i.e., $\mathcal{S}$ is not equal to $\mathcal{V}$). What happens if this is not the case?

Exercise 2. In this exercise, we will see some properties of orthogonal complements. Recall that for a subspace $\mathcal{S}$ of the vector space $\mathcal{V} \subseteq \mathbb{R}^n$, we define the orthogonal complement of $\mathcal{S}$ as

$$\mathcal{S}_\mathcal{V}^\perp = \{\mathbf{v} \in \mathcal{V} : \mathbf{v} \perp \mathcal{S}\} = \{\mathbf{v} \in \mathcal{V} : \mathbf{v} \perp \mathbf{s} \forall \mathbf{s} \in \mathcal{S}\}.$$

- (a) Show that $\mathcal{S}_\mathcal{V}^\perp$ is a vector space.

Hint: Can you use the subspace trick?

- (b) Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two subspaces of $\mathcal{V} \subset \mathbb{R}^n$ such that $\mathcal{S}_1 \subseteq \mathcal{S}_2$. Then, show that $\mathcal{S}_{1,\mathcal{V}}^\perp \supseteq \mathcal{S}_{2,\mathcal{V}}^\perp$.
- (c) Show that $(\mathcal{S}_\mathcal{V}^\perp)^\perp = \mathcal{S}$. Hence, $\mathcal{S}_1 \subseteq \mathcal{S}_2$ if and only if $\mathcal{S}_{1,\mathcal{V}}^\perp \supseteq \mathcal{S}_{2,\mathcal{V}}^\perp$.

Note: Go back to the previous exercise. Observe that $\{\mathbf{u}_{p+1}, \dots, \mathbf{u}_{p+q}\}$ constructed in that exercise is a basis for $\mathcal{S}_\mathcal{V}^\perp$. Thus,

$$\dim(\mathcal{S}) + \dim(\mathcal{S}_\mathcal{V}^\perp) = \dim(\mathcal{V}).$$

Note: For a subspace $\mathcal{S}$ of n -vectors (i.e., $\mathcal{S}$ contains tuples of size n), we usually denote its orthogonal complement as $\mathcal{S}^\perp$, instead of $\mathcal{S}_{\mathbb{R}^n}^\perp$. Then, it follows that $\dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) = n$.

- (d) Find $\mathcal{S}^\perp$ for the following subspaces.

i. $\mathcal{S} = \mathcal{C}((1, 0)^\top)$.

- ii. $\mathcal{S} = \mathcal{C}((1, 0, 0)^\top)$.
- iii. $\mathcal{S} = \mathcal{C}\left(\begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}\right)$.
- iv. $\mathcal{S} = \mathcal{C}\left(\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}\right)$.
- v. $\mathcal{S} = \mathcal{C}\left(\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}\right)$.

Exercise 3. Observe that the result in Exercise 1 can be rewritten as

$$\mathcal{V} = \{\mathbf{w} + \mathbf{z} : \mathbf{w} \in \mathcal{S}, \mathbf{z} \in \mathcal{S}_\mathcal{V}^\perp\}.$$

Given two vector subspaces $\mathcal{S}$ and $\mathcal{T}$ of $\mathbb{R}^n$, we can always define

$$\mathcal{S} + \mathcal{T} := \{\mathbf{s} + \mathbf{t} : \mathbf{s} \in \mathcal{S}, \mathbf{t} \in \mathcal{T}\}.$$

- (a) Show that $\mathcal{S} + \mathcal{T}$ is a vector space. This is called the **sum** of $\mathcal{S}$ and $\mathcal{T}$.
- (b) Show that $\mathcal{S} \cup \mathcal{T} \subseteq \mathcal{S} + \mathcal{T}$.
- (c) Show that if $\mathcal{V}$ is any vector space such that $\mathcal{V} \supseteq \mathcal{S} \cup \mathcal{T}$, then $\mathcal{V} \supseteq \mathcal{S} + \mathcal{T}$.

Note: In general, for vector spaces $\mathcal{S}$ and $\mathcal{T}$, $\mathcal{S} \cup \mathcal{T}$ is NOT a vector space (why?). Parts (b) and (c) show that $\mathcal{S} + \mathcal{T}$ is the smallest vector space containing $\mathcal{S} \cup \mathcal{T}$.

Note: For two subspaces $\mathcal{S}$ and $\mathcal{T}$ of $\mathbb{R}^n$, we call $\mathcal{S} + \mathcal{T}$ the **direct sum** if every vector $\mathbf{v} \in \mathcal{S} + \mathcal{T}$ has a **unique** representation $\mathbf{v} = \mathbf{s} + \mathbf{t}$, where $\mathbf{s} \in \mathcal{S}, \mathbf{t} \in \mathcal{T}$. We denote the direct sum by $\mathcal{S} \oplus \mathcal{T}$.

- (d) Show that $\mathcal{S} + \mathcal{T}$ is the direct sum if and only if the elements of $\mathcal{S}$ and $\mathcal{T}$ are mutually linearly independent. That is, for $\mathbf{s}(\neq \mathbf{0}) \in \mathcal{S}$ and $\mathbf{t}(\neq \mathbf{0}) \in \mathcal{T}$, $\mathbf{s}$ and $\mathbf{t}$ are linearly independent. This is equivalent to saying that a basis for $\mathcal{S}$ is linearly independent to a basis for $\mathcal{T}$.
- (e) $\mathcal{S}$ and $\mathcal{T}$ are said to be orthogonal to each other if $\mathbf{s}^\top \mathbf{t} = 0$ for all $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$. We denote this by writing $\mathcal{S} \perp \mathcal{T}$. Show that if $\mathcal{S} \perp \mathcal{T}$, then $\mathcal{S} + \mathcal{T} = \mathcal{S} \oplus \mathcal{T}$.

Note: By definition, $\mathcal{S}$ and $\mathcal{S}_\mathcal{V}^\perp$ are orthogonal to each other. The result in Exercise 1 can now be written as $\mathcal{V} = \mathcal{S} \oplus \mathcal{S}_\mathcal{V}^\perp$. In particular, for a subspace $\mathcal{S}$ of $\mathbb{R}^n$, $\mathcal{S} \oplus \mathcal{S}^\perp = \mathbb{R}^n$.

- (f) Show that $\mathcal{S} \cap \mathcal{T}$ is always a vector space.
- (g) (**Hard**) Show that $\dim(\mathcal{S} + \mathcal{T}) + \dim(\mathcal{S} \cap \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T})$.

Hint: Try to construct bases for each of the spaces $\mathcal{S}, \mathcal{T}, \mathcal{S} \cap \mathcal{T}$ and $\mathcal{S} + \mathcal{T}$ with some common elements.

Note: Verify the result for the direct sum. What is $\mathcal{S} \cap \mathcal{T}$ when $\mathcal{S}$ and $\mathcal{T}$ are mutually linearly independent?

- (h) Let $\mathbf{X}_{n \times p}$ and $\mathbf{Z}_{n \times q}$ be two matrices. Let $[\mathbf{X}, \mathbf{Z}]_{n \times (p+q)}$ be the matrix obtained by placing $\mathbf{X}$ and $\mathbf{Z}$ next to each other. Show that $\mathcal{C}([\mathbf{X}, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{Z})$. Moreover, if the columns of $\mathbf{X}$ and the columns of $\mathbf{Z}$ are mutually orthogonal, then $\mathcal{C}([\mathbf{X}, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{Z})$.

Note: The columns of $\mathbf{X}$ and $\mathbf{Z}$ are mutually orthogonal if and only if $\mathbf{X}^\top \mathbf{Z} = \mathbf{O}$, where $\mathbf{O}$ is a matrix whose all entries are 0.

(i) Find $\mathcal{S} \cap \mathcal{T}$, $\mathcal{S} \cup \mathcal{T}$ and $\mathcal{S} + \mathcal{T}$ in the following cases. Check if $\mathcal{S} \cup \mathcal{T}$ is a vector space. Also verify (g) above. Is $\mathcal{S} + \mathcal{T}$ a direct sum?

i. $\mathcal{S} = \mathcal{C}((1, 0)^\top)$ and $\mathcal{T} = \mathcal{C}((0, 1)^\top)$.

ii. $\mathcal{S} = \mathcal{C}((1, 0)^\top)$ and $\mathcal{T} = \mathcal{C}((1, 1)^\top)$.

iii. $\mathcal{S} = \mathcal{C} \left(\begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix} \right)$ and $\mathcal{T} = \mathcal{C} \left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \right)$.

iv. $\mathcal{S} = \mathcal{C} \left(\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \right)$ and $\mathcal{T} = \mathcal{C} \left(\begin{bmatrix} 2 \\ 4 \\ 9 \end{bmatrix} \right)$.

Exercise 4. Here is a generalization of the sum and direct sum for more than two vector spaces. Let $\mathcal{S}_1, \dots, \mathcal{S}_m$ be vector subspaces of $\mathbb{R}^n$. We define

$$\mathcal{S}_1 + \dots + \mathcal{S}_m = \{\mathbf{s}_1 + \dots + \mathbf{s}_m : \mathbf{s}_1 \in \mathcal{S}_1, \dots, \mathbf{s}_m \in \mathcal{S}_m\}$$

to be the **sum** of $\mathcal{S}_1, \dots, \mathcal{S}_m$.

(a) Show that $\mathcal{S}_1 + \dots + \mathcal{S}_m$ is the smallest vector space containing $\mathcal{S}_1 \cup \dots \cup \mathcal{S}_m$.

(b) Show that $\dim(\mathcal{S}_1 + \dots + \mathcal{S}_m) \leq \dim(\mathcal{S}_1) + \dots + \dim(\mathcal{S}_m)$.

If every vector $\mathbf{v} \in \mathcal{S}_1 + \dots + \mathcal{S}_m$ has a **unique** representation $\mathbf{v} = \mathbf{s}_1 + \dots + \mathbf{s}_m$, where $\mathbf{s}_i \in \mathcal{S}_i, i = 1, \dots, m$, then $\mathcal{S}_1 + \dots + \mathcal{S}_m$ is called the **direct sum** of $\mathcal{S}_1, \dots, \mathcal{S}_m$, and it is denoted by $\mathcal{S}_1 \oplus \dots \oplus \mathcal{S}_m$.

(c) Verify that if $\mathcal{S}_1, \dots, \mathcal{S}_m$ are mutually orthogonal, i.e., $\mathcal{S}_i \perp \mathcal{S}_j \forall i \neq j$, then $\mathcal{S}_1 + \dots + \mathcal{S}_m = \mathcal{S}_1 \oplus \dots \oplus \mathcal{S}_m$.

(d) Show that $\dim(\mathcal{S}_1 \oplus \dots \oplus \mathcal{S}_m) = \dim(\mathcal{S}_1) + \dots + \dim(\mathcal{S}_m)$.

Exercise 5. In this exercise, we will study some properties of column spaces of matrices. You can also use these as a collection of results. In the following, A, B, C etc. are matrices of appropriate dimensions. The dimension is not always explicitly defined; determine them before proceeding. Prove the following.

(a) $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$.

(b) $\mathcal{C}(AA^\top) = \mathcal{C}(A)$. Hence, $\text{rank}(AA^\top) = \text{rank}(A)$.

(c) $\mathcal{C}(A) \subseteq \mathcal{C}(B)$ if and only if $A = BC$ for a suitable matrix C.

Note: What can you say about the relationship between the ranks of A and B? What about the relationship between the number of columns of A and B?

(d) $A^\top B = \mathbf{O}$ if and only if $\mathcal{C}(A) \subseteq \mathcal{C}(B)^\perp$.

(e) If $\mathcal{C}(A) \subseteq \mathcal{C}(B)$ and $\text{rank}(A) = \text{rank}(B)$, then $\mathcal{C}(A) = \mathcal{C}(B)$.

(f) $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.

(g) $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

Exercise 6. Let $A_{n \times p}$ be a matrix. We define the **null space** of A to be

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^p : A\mathbf{x} = \mathbf{0}\}.$$

Recall that we can view the matrix A as a linear map between $\mathbb{R}^p$ and $\mathbb{R}^n$. $\text{Null}(A)$ is the collection of vectors that are mapped to $\mathbf{0}$.

- (a) Show that $\text{Null}(A)$ is a subspace of $\mathbb{R}^p$.
- (b) Show that $\text{Null}(A) = \mathcal{C}(A^\top)^\perp$. Equivalently, $\text{Null}(A) = \mathcal{R}(A)^\perp$.
- (c) Use part (b) to show that $\dim(\text{Null}(A)) = p - \text{rank}(A)$.

Note: The dimension of $\text{Null}(A)$ is called the **nullity** of A . Part (c) shows that $\text{nullity}(A) = p - \text{rank}(A)$. Put differently, $\text{rank}(A) + \text{nullity}(A) = p$, the number of columns of A . This is also called the **rank-nullity theorem**.

Note: If $\mathbf{b}_0$ is a solution to $A\mathbf{b} = \mathbf{c}$, then so is $\mathbf{b}_0 + \mathbf{x}$ for any $\mathbf{x} \in \text{Null}(A)$. What is the connection between the null space and the uniqueness of the solution?

- (d) Find the null spaces of the matrices in Exercise 2 part (d). Verify properties (b) and (c) above.

Exercise 7. We saw in the class that for any subspace $\mathcal{S} \subseteq \mathbb{R}^n$, there exists a projection matrix onto $\mathcal{S}$. Moreover, we asserted that such a matrix is unique if we restrict it to be symmetric. We call such a matrix the **orthogonal projection** onto $\mathcal{S}$ and denote it by $P_{\mathcal{S}}$.

- (a) Verify the assertion: If $\mathcal{S}$ is a subspace of $\mathbb{R}^n$ and P, Q are symmetric projection matrices onto $\mathcal{S}$, then $P = Q$.

Hint: For any $\mathbf{v} \in \mathbb{R}^n$, $P\mathbf{v} \in \mathcal{S}$. What can be said about $QP\mathbf{v}$?

- (b) Let $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ be an orthonormal basis of $\mathcal{S}$. Show that the orthogonal projection onto $\mathcal{S}$ is given by $P_{\mathcal{S}} := \sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i^\top$.

Hint: Verify the properties of an orthogonal projection matrix, viz., $P^2 = P$, $P\mathbf{v} \in \mathcal{S}$ for every $\mathbf{v} \in \mathbb{R}^n$ and $P^\top = P$.

From part (a), it follows that $\sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i^\top$ DOES NOT depend on the choice of the orthonormal basis. In the following, we verify this differently.

- (c) Show that $\sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i^\top = UU^\top$, where $U = (\mathbf{u}_1, \dots, \mathbf{u}_r)$. Observe that the columns of U are orthonormal, so that U is an orthogonal matrix.
- (d) Consider a different orthonormal basis $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ of $\mathcal{S}$. Define $V = (\mathbf{v}_1, \dots, \mathbf{v}_r)$. Show that there exists a square matrix W such that $U = VW$.
- (e) Show that W in the previous step satisfies $W^\top W = I_r$.

Hint: The columns of U are orthonormal. The same is true for V .

- (f) Show that the previous part implies W is invertible and $W^{-1} = W^\top$.
- (g) Show that $UU^\top = VV^\top$.

Exercise 8. Here we check some facts about orthogonal projections.

- (a) Show that the only non-singular orthogonal projection matrix is the identity matrix I . Hence, any orthogonal projection that is not I must be singular. In particular, orthogonal projections onto proper subspaces are **rank deficit** (not of full rank).

(b) Let $\mathcal{S}_1, \mathcal{S}_2$ be two subspaces of $\mathbb{R}^n$. Recall that $\mathcal{S}_1$ and $\mathcal{S}_2$ are said to be orthogonal, denoted $\mathcal{S}_1 \perp \mathcal{S}_2$, if $\mathbf{s}_1 \perp \mathbf{s}_2$ for every $\mathbf{s}_1 \in \mathcal{S}_1, \mathbf{s}_2 \in \mathcal{S}_2$. Show that $\mathcal{S}_1 \perp \mathcal{S}_2$ if and only if $P_{\mathcal{S}_1}P_{\mathcal{S}_2} = \mathbf{O}$.

(c) Show that $\mathcal{S}_1 \subseteq \mathcal{S}_2$ if and only if $P_{\mathcal{S}_1}P_{\mathcal{S}_2} = P_{\mathcal{S}_2}P_{\mathcal{S}_1} = P_{\mathcal{S}_1}$.

Note: For two subspaces $\mathcal{S}_1$ and $\mathcal{S}_2$, $P_{\mathcal{S}_2}P_{\mathcal{S}_1}\mathbf{x}$ is the result of first projecting $\mathbf{x}$ onto $\mathcal{S}_1$ and then projecting the result onto $\mathcal{S}_2$. In general, $P_{\mathcal{S}_2}P_{\mathcal{S}_1}$ and $P_{\mathcal{S}_1}P_{\mathcal{S}_2}$ need not be the same.

(d) Show that if $\mathcal{S}_1$ and $\mathcal{S}_2$ are orthogonal subspaces of $\mathbb{R}^n$, then $P_{\mathcal{S}_1 \oplus \mathcal{S}_2} = P_{\mathcal{S}_1} + P_{\mathcal{S}_2}$.

Note: Is orthogonality of $\mathcal{S}_1$ and $\mathcal{S}_2$ necessary for this result?

(e) Show that if $\mathcal{S}_1 \subseteq \mathcal{S}_2$, then $P_{\mathcal{S}_2} - P_{\mathcal{S}_1}$ is also an orthogonal projection. So, the difference of two orthogonal projections can be an orthogonal projection. Where does it project onto?

(f) Let $X^{n \times p} = (X_1^{n \times q}, X_2^{n \times r})$ be of full column rank. Show that $X(X^\top X)^{-1}X^\top - X_1(X_1^\top X_1)^{-1}X_1^\top$ is an orthogonal projection. Where does it project onto?

Note: Observe that X having full column rank implies both $X^\top X$ and $X_1^\top X_1$ are invertible.

Exercise 9. When dealing with column spaces of a matrix A , i.e., when $\mathcal{S} = \mathcal{C}(A)$, we will write P_A (instead of $P_{\mathcal{C}(A)}$) to denote the orthogonal projection onto $\mathcal{C}(A)$. Let A and B be two matrices with the same number of rows. We write $[A, B]$ to denote the matrix obtained by stacking A and B side-by-side.

(a) Show that $\mathcal{C}(A) \perp \mathcal{C}((I - P_A)B)$.

(b) Show that $\mathcal{C}(A) + \mathcal{C}((I - P_A)B) = \mathcal{C}([A, B])$. Conclude that $\mathcal{C}([A, B]) = \mathcal{C}(A) \oplus \mathcal{C}((I - P_A)B)$.

(c) Show that $P_{[A, B]} = P_A + P_{(I - P_A)B}$.

Note: The last result can be seen as orthogonalization for subspaces. Observe the similarity with the Gram-Schmidt procedure.